SECTIONS-WITH-SUPPORT FUNCTORS COMMUTE WITH DIRECT SUMS OVER A RETROCOMPACT CLOSED SUBSET

THE STACKS PROJECT AUTHORS

Editorial note. The source marks the functor-composition commutation '(details omitted)'; carried as an explicit obligation.

Editorial note. Input lemma-quasi-coherence-pushforward-direct-sums is same-chapter NOT carded (cross-line-claimed label) and is declared external.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 1086 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Triangle for complexes with support. Cohomology, lemma triangle-sections-with-support-sheaves (cross-chapter, named). Expresses i_*R H_T(E) as the fibre of E -> Rj_*(E|_U).
- Pushforward with supports commutes with direct sums. lemma-quasi-coherence-pushforward-direct-sums (same chapter, NOT carded, declared external). Rj_*(-) commutes with direct sums on D_QCoh for retrocompact U.
- Equivalence for complexes supported on a closed subset. Cohomology, lemma complexes-with-support-on-closed (cross-chapter, named). i_* identifies D(i^{-1}O_X) with the derived category of complexes supported on T.

Source reference. Original reference 0AEF: focused-seeds/geometry/S9-perf2-support-direct-sums/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0G72: focused-seeds/geometry/S9-perf2-support-direct-sums/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08DZ: sources/stacks-perfect/pinned-original-source.tex, source lines 803–803. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let X be a scheme. Let $T \subset X$ be a closed subset such that $X \setminus T$ is a retrocompact open of X. Then for a family of objects E_i, $i \in I$ of $D_{QCoh}(\mathcal{O}_X)$ we have $R\mathcal{H}_T(\bigoplus E_i) = \bigoplus R\mathcal{H}_T(E_i)$.*

Proof. Set $U = X \setminus T$ and denote $j : U \to X$ the inclusion. By Cohomology, Lemma 0G72 there is a distinguished triangle

$$i_* R\mathcal{H}_T(E) \to E \to Rj_*(E|_U) \to i_* R\mathcal{H}_Z(E)[1]$$

in $D(\mathcal{O}_X)$ for any E in $D(\mathcal{O}_X)$. The functor $E \mapsto Rj_*(E|_U)$ commutes with direct sums on $D_{QCoh}(\mathcal{O}_X)$ by Lemma 08DZ. It follows that the same is true for the functor $i_* \circ R\mathcal{H}_T$ (details omitted). Since $i_* : D(i^{-1}\mathcal{O}_X) \to D_T(\mathcal{O}_X)$ is an equivalence (Cohomology, Lemma 0AEF) we conclude. □